

Employee Attrition Prediction & HR Analytics System

Project Report | IBM HR Analytics dataset (1,470 employees) | Python, scikit-learn, SHAP

1. Introduction

Employee attrition is costly: it raises recruitment and training expenses, disrupts teams and erodes institutional knowledge. Most HR teams react only after a resignation. This project takes a predictive approach, using historical HR records to identify which employees are at risk of leaving and, just as importantly, *why*. The aim is to give HR decision-makers early, explainable signals so retention actions can be targeted rather than generic.

2. Abstract

The project analyses the IBM HR Analytics dataset (1,470 employees, 35 attributes) with an overall attrition rate of 16.12%. After cleaning and exploratory analysis, three classifiers (Logistic Regression, Decision Tree and Random Forest) were trained on an 80/20 stratified split with class-weight balancing to handle the imbalance between leavers and stayers. Logistic Regression performed best (ROC-AUC 0.807, recall 76.6%), followed by Random Forest (0.778) and Decision Tree (0.638). SHAP explainability on the Random Forest identified OverTime, Age, Monthly Income, Stock Option Level and Years at Company as the strongest drivers of attrition risk. EDA showed that employees working overtime leave at roughly three times the rate of those who do not (30.5% vs 10.4%), and that Sales Representatives have the highest role-level attrition (39.8%).

3. Tools Used

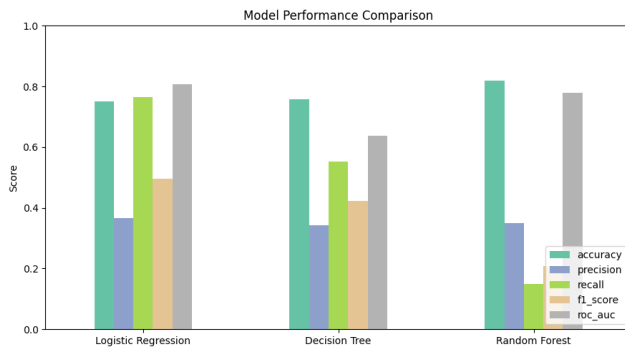
Category	Tools / Libraries	Purpose
Language	Python 3	End-to-end scripting of the pipeline (three modular scripts)
Data handling	pandas, NumPy	Loading, cleaning, encoding and aggregation
Visualisation	Matplotlib, Seaborn	EDA charts, confusion matrices, ROC curves, comparison plots
Machine learning	scikit-learn	Logistic Regression, Decision Tree, Random Forest, scaling, metrics
Explainability	SHAP (TreeExplainer)	Global feature impact and individual-employee explanations
Persistence / BI	joblib, CSV export	Saving trained model and artefacts; cleaned CSV ready for Power BI / Tableau / Excel

4. Steps Involved in Building the Project

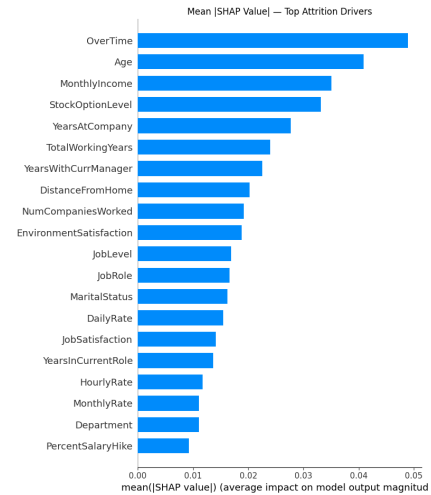
- Data loading and cleaning (01_eda.py).** Loaded the raw CSV (1,470 rows, 35 columns). Dropped four non-informative columns (EmployeeCount, Over18, StandardHours, EmployeeNumber), checked for missing values and duplicates, and encoded the target (Attrition: Yes = 1, No = 0). The cleaned 31-column file was saved for modelling and BI use.
- Exploratory data analysis.** Generated 11 charts covering attrition by department, job role, overtime, income, job satisfaction, work-life balance, tenure, promotion history, a correlation heatmap and salary bands. Key finding: attrition is concentrated among overtime workers, low-income and early-tenure employees, and Sales roles.
- Feature preparation (02_modeling.py).** Label-encoded categorical variables, performed a stratified 80/20 train-test split (random_state = 42) and standardised features for Logistic Regression.
- Model training.** Trained Logistic Regression, a depth-6 Decision Tree and a 300-tree Random Forest (max depth 8), all with class_weight = "balanced" to counter the 16% minority class.
- Evaluation.** Compared accuracy, precision, recall, F1 and ROC-AUC, and produced confusion matrices, ROC curves and Random Forest feature importances. Recall and ROC-AUC were prioritised, since missing a leaver costs more than a false alarm.
- Explainability (03_shap_explainability.py).** Applied SHAP TreeExplainer to the Random Forest to produce a global importance bar chart, a beeswarm summary and waterfall plots for one high-risk and one low-risk employee.

Results at a Glance

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.752	0.367	0.766	0.497	0.807
Decision Tree	0.759	0.342	0.553	0.423	0.638
Random Forest	0.820	0.350	0.149	0.209	0.778



Model performance comparison



SHAP global feature importance (Random Forest)

5. Conclusion

The project shows that attrition can be predicted and explained from routine HR data. Logistic Regression gave the best balance of recall and ROC-AUC, catching about 77% of actual leavers on the test set, though its precision (about 37%) means HR would review a number of false alarms. Random Forest achieved the highest accuracy but low recall (15%), so it is less suitable for flagging leavers at the default threshold. SHAP showed the drivers are actionable: overtime burden, compensation and stock options, age and tenure. Recommended actions are to review overtime and workload, strengthen compensation and equity for early-tenure staff, and prioritise retention programmes in Sales. Future work includes threshold tuning, cross-validation and hyperparameter search, class-imbalance techniques such as SMOTE, and deploying the model as a live risk-scoring dashboard. Note that results come from a single, publicly available dataset with a 294-employee test set, so they should be validated on an organisation's own data before use.